Download and Preprocessing

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This script is used to download and preprocess the raw sequencing data from the RNA-seq experiments of the Ageing hallmarks exhibit organ-specific temporal signatures. Here, we focus on the male samples, particularly on the bone tissue at 1 and 24 age stages:

0.1 Download data

To access the data download, we use the program `fasterq-dump`. This is already installed in the Chaac server, so we can use subprocess from Python in order to make the download. To do so, we need the metadata for the experiment; this is stored in the GEO accession number GSE132040.

```
#| eval: false

#Download the metadata table
wget https://ftp.ncbi.nlm.nih.gov/geo/series/GSE132nnn/GSE132040/suppl/ \
GSE132040%5FMACA%5FBulk%5Fmetadata.csv.gz -O data/Tabula_Muris_GSE_metadata.csv.gz

gunzip data/Tabula_Muris_GSE_metadata.csv.gz
```

```
The columns present in the metadata are:

- Sample name
- title
- source name
- organism
- characteristics: age
- characteristics: developmental stage
- characteristics: sex
- molecule
- description
- processed data file
- raw file
- BioSample
- Instrument Model
The organims from who the samples come form is ['Mus musculus C57/BL6']
The SRR id for the experiment C20_384Bulk_Plate1_S68 is SRR9126934
The SRR id for the experiment C20_384Bulk_Plate2_S68 is SRR9127010
The SRR id for the experiment D6_384Bulk_Plate2_S78 is SRR9127296
The SRR id for the experiment E3_384Bulk_Plate3_S99 is SRR9126859
The SRR id for the experiment F21_384Bulk_Plate1_S141 is SRR9127157
The SRR id for the experiment F7_384Bulk_Plate3_S127 is SRR9127369
The SRR id for the experiment J7_384Bulk_Plate1_S223 is SRR9127396
The SRR id for the experiment O9_384Bulk_Plate1_S345 is SRR9127293
So the SRR ids are:

- SRR9126934
- SRR9127010
```

```
- SRR9127296
- SRR9126859
- SRR9127157
- SRR9127369
- SRR9127396
- SRR9127293
```

With the IDs now we can download them

```
for srr in SRR_boneM:
    subP.run(["qsub", "-N", f"Download_{srr}", "src/jdl/Download.jdl", srr])
```

The downloaded files are in `fastq` format. This is the general file format for short sequence data. The characteristics of this kind of file are that it not only contains the nucleotide sequence resulting from the sequencing reaction, but also has more properties such as the corresponding score for every nucleotide that is sequenced and the IDs for each sequence, having a total of 4 lines for each read sequenced. With this information provided by this format, we can deliver a FastQC (quality control) report with the objective of having the following statistics of our sequences:

- **Per Base sequence quality**
- **Per tile sequence quality**
- **Per sequence quality scores**
- **Per base sequence content**
- **Per sequence GC content**
- **Per base N content**
- **Sequence Duplication Levels**

This information in the reports is critical not only for knowing if the sequencing reaction procedure was done correctly, but also helps us to make decisions like trimming those bases that could have a low quality at the end or start of our sequences (as it frequently occurs with Illumina sequences).

```
# | eval: false

#make the fastqc reports
qsub -N fastqc_report src/jdl/fastqc_rep.jdl data/
```

With the analysis product of the FastQC and MultiQC reports, we can make decisions about the preprocessing of the data:

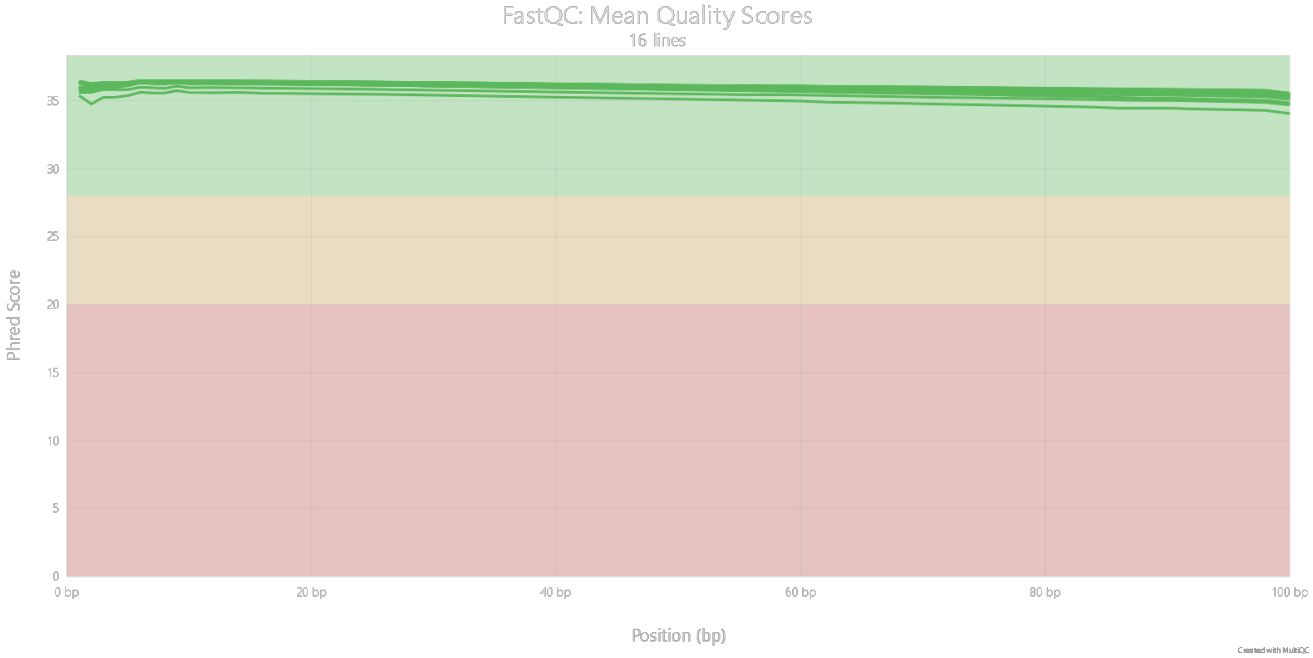


Figure 1: fastqc_per_base_sequence_quality_plot.png

After reviewing all the FastQC reports in the HTML view, we can see that the statistics related to the base quality and the sequence length are constant over all the samples and with good results, without having neither low scores nor inconstant sequence lengths. These results are characteristic of current NGS.

0.2 Possible problems

0.2.1 Base content

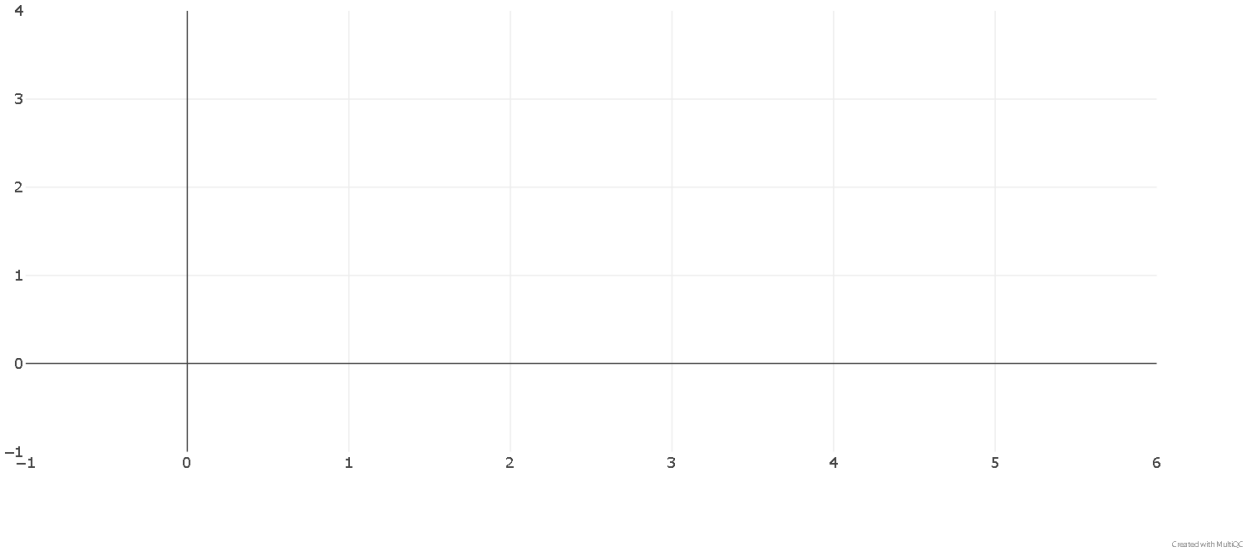


Figure 2: fastqc_per_base_sequence_content_plot.png

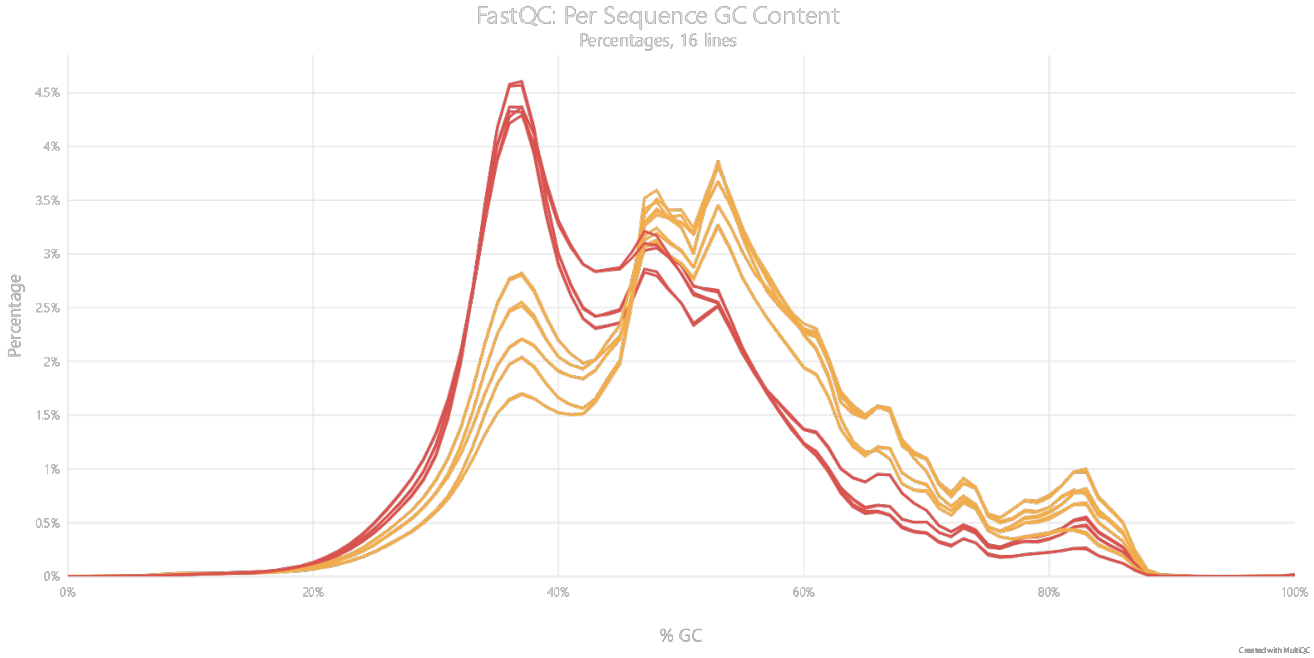


Figure 3: fastqc_per_sequence_gc_content_plot.png

One of the first problems we can see across all the samples is the base content related to the first bases in the reads. We can see that the base sequence content is very inconstant, having no proportional content of bases in a 1-15 range. As this problem is constant in all the libraries, it can be solved by trimming the sequences in their first base pairs.

```
# | eval: false

#first iterarte over de data directory
for srr_dir in data/*/; do
    #extract the SRR id
    srr_id="${srr_dir%/*}"
    srr_id="${srr_id##*/}"
    echo "Procesing the $srr_id fastq files"
    #iterate over the files that are in the SRR directory
    for srr in "$srr_dir"*.fastq; do
        #now we check if it is the paired end file
        if [[ $srr == *_1.fastq* ]];then
            #we assume taht it has it paired file so we can obtain the id
            paired_1=$srr
            paired_2=${paired_1/_1.fastq/_2.fastq}
            echo "Files to be proccesed $paired_1 and $paired_2"
            qsub -N "$srr_id"_cleaning src/jdl/fastp_cleaning.jdl $paired_1 $paired_2
            break
        else
            #we are proccesing a unpaired-end file
            qsub -N "$srr_id"_cleaning src/jdl/fastp_cleaning.jdl $srr
        fi
    done
done
```

0.2.2 Duplication and overrepresented sequences

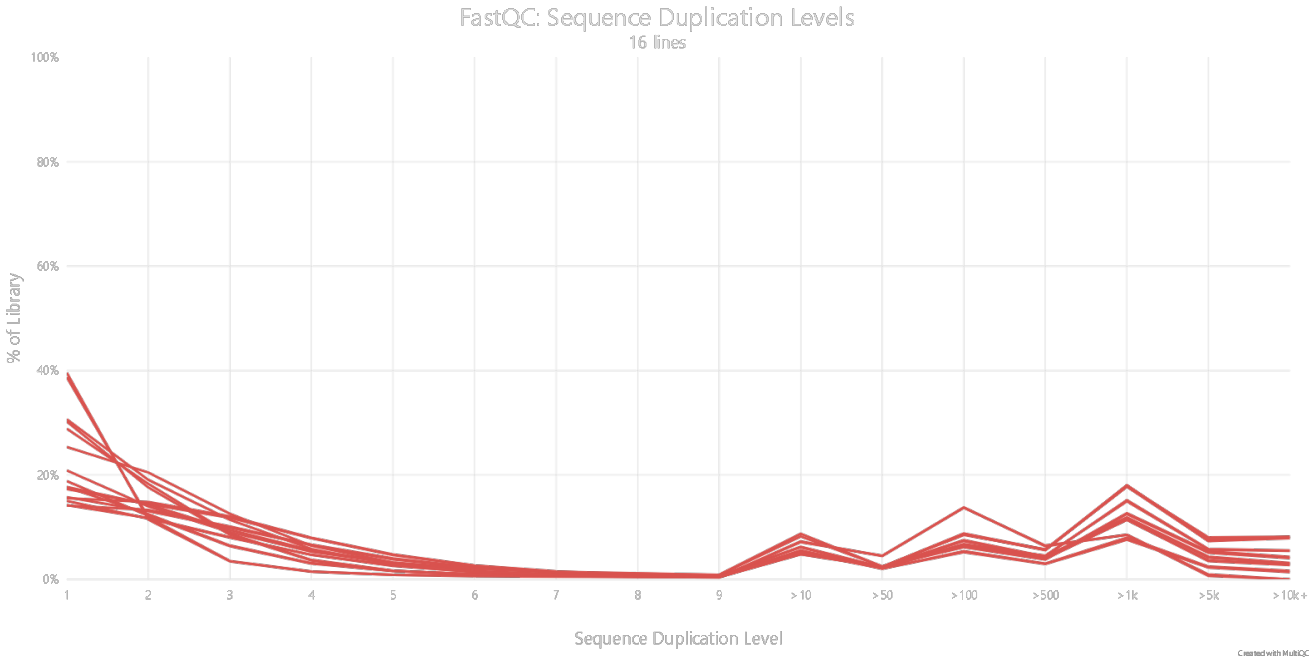
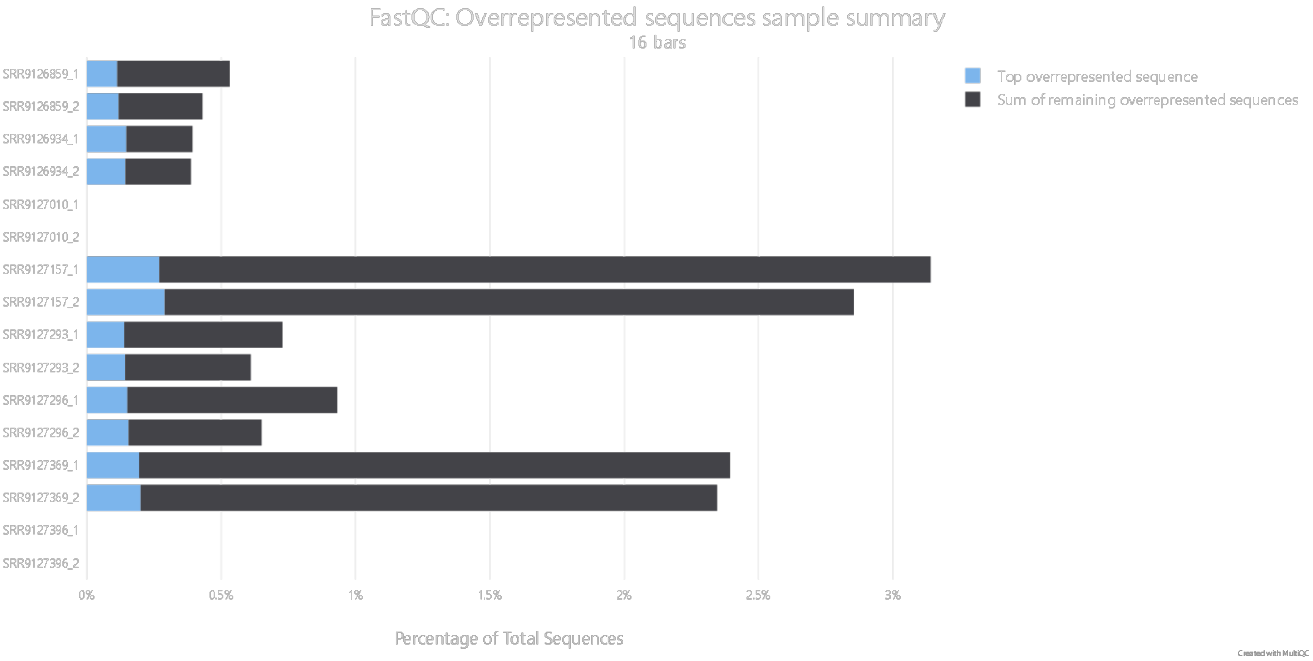
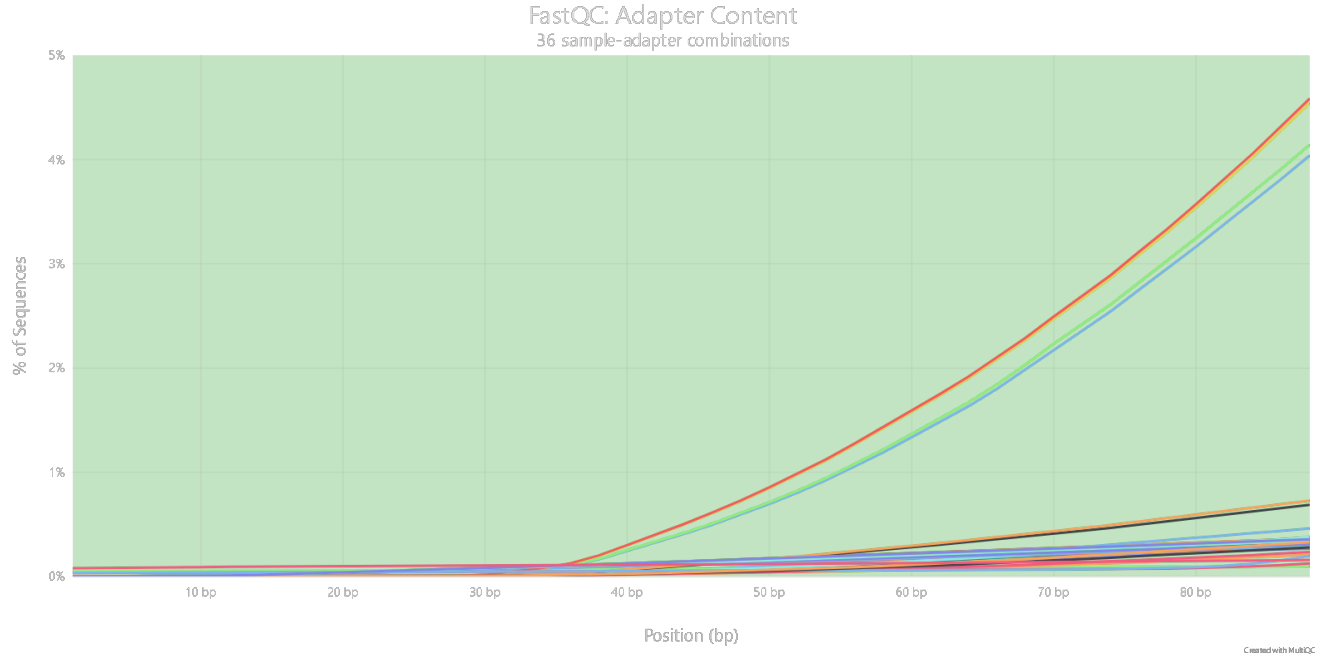


Figure 4: fastqc_sequence_duplication_levels_plot.png

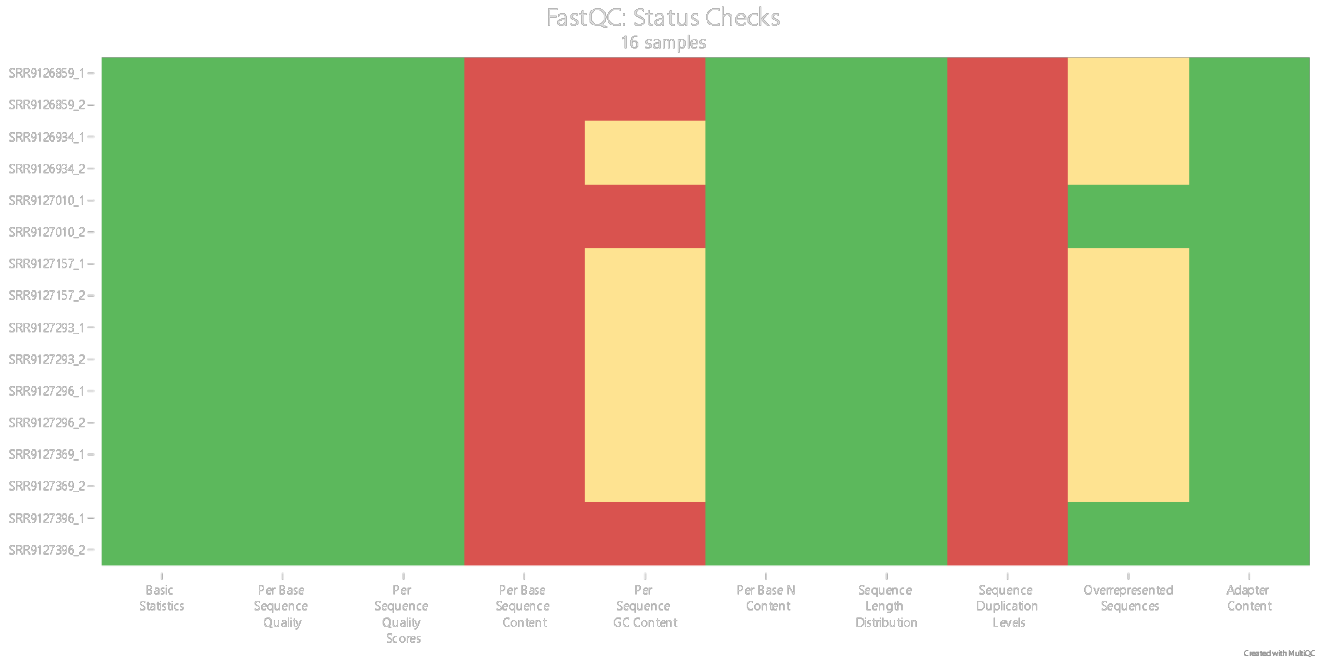


Another warning that all our samples have is the duplication level of sequences. We can see that this statistic can reach 72% of all the reads in some samples. Also, the majority of the SRR files have overrepresented sequences, with only those from the IDs SRR9127396 and SRR9127010 being the ones that do not have this problem.



In general, all the samples do not have a significant adapter content, with only two having a little representation of Nextera transposase sequence (SRR9126859 and SRR9127369). Despite this, it does not arise a warning.

0.3 General results



In this heatmap representing the problems across the SRR files, we can see that all the warnings are consistent across all the samples with few exceptions that we discussed above. So this gives us the first insight that our data is comparable, obviously with the precise normalization methods and preprocessing ones.

Now we make another time the QC report with the data processed by fastp.

```
# | eval: false

#make the fastqc reports
qsub -N fastqc_report src/jdl/fastqc_rep.jdl data clean.fastq results/fastqc/fastp
```

We can see in the FastQC reports that the new trimmed sequences have improved at the start of all sequences after removing the first 15 base pairs of all our SRR files.

Once we have the files processed, we can start with the alignments of the data, so we finally save the relevant information such as the metadata table of our specific samples.

`The metadata tables looks:`

Table 1

src	name	months	srr_id	biosample	plate
C20_384Bulk_Plate1_S68	Bone_8	24	SRR9126934	SAMN11854715	Plate1
C20_384Bulk_Plate2_S68	Bone_16	1	SRR9127010	SAMN11854716	Plate2
D6_384Bulk_Plate2_S78	Bone_12	1	SRR9127296	SAMN11854792	Plate2
E3_384Bulk_Plate3_S99	Bone_21	24	SRR9126859	SAMN11854844	Plate3
F21_384Bulk_Plate1_S141	Bone_46	1	SRR9127157	SAMN11854896	Plate1
F7_384Bulk_Plate3_S127	Bone_32	1	SRR9127369	SAMN11854915	Plate3
J7_384Bulk_Plate1_S223	Bone_11	24	SRR9127396	SAMN11855147	Plate1
O9_384Bulk_Plate1_S345	Bone_35	24	SRR9127293	SAMN11855451	Plate1

Now we this pre-proccesed data and the metada data allready saved we can start with the next step, the same that is align the reads to a reference genome or in other words make all the reads sequenced to have sense in biological terms in the form that they have identification as places in the reference genome of the specie they came from.